

MATH

Majoring & Minorin

The department offers the following three majors leading to either the BA or BS degree: Mathematics, Computer Science and Computational Mathematics. Each student's major program must be determined in consultation with the student's academic advisor; the program should reflect the student's personal needs and goals. In particular, students intending to study mathematics or computer science at the graduate level are encouraged to take additional upper-level electives beyond those satisfying the minimum requirements for the major. The Computational Mathematics major is designed to provide students with a foundational mastery of the interdependent disciplines of Mathematics and Computer Science. The curriculum is a blend of core courses intended to provide a broad knowledge base while maintaining depth in both subject areas. The department also offers a minor in Mathematics and in Computer Science.

All majors in the department are required to successfully complete the designated senior seminar in their respective majors or to carry out a department-approved senior project to satisfy the capstone experience requirement. All majors must successfully pass the Major Field Test in Mathematics or Computer Science.

Notes: A minimum grade of C- must be earned in any course if it is to count as a prerequisite for a subsequent Mathematics or Computer Science course. Computer Science 54-184, 54-284, and 54-454 cannot be taken out of sequence without departmental approval. Any three-credit or four-credit course at the 300 level or above will serve as an approved course unless the department explicitly indicates otherwise. Majors are recommended to start taking approved upper-level courses no later than their junior year. Credit for Honors (Mathematics 52-984 or Computer Science 54-984) will be counted, at most, once as an approved upper-level elective.

MATHEMATICS

COMPUTATIONAL MATHEMATICS

ASSOCIATED MAJORS/MINORS

Major in Computational Mathematics (BA): 14 courses

- CSC54-184 Computer Science I
- CSC54-284 Computer Science II
- CSC54-384 Discrete Mathematics
- CSC54-394 Computer Organization
- CSC54-454 Algorithms
- MAT52-164 Modern Calculus I
- MAT52-264 Modern Calculus II
- MAT52-364 Modern Calculus III
- MAT52-674 Linear Algebra
- MAT52-754 Differential Equations
- Two courses from:
 - CSC54-414 Operations Research
 - CSC54-514 Database Management
 - MAT52-524 Introduction to Numerical Analysis
 - MAT52-574 Probability and Mathematical Statistics
- One additional approved upper-level course, in Mathematics or Computer Science or PHY53-454 Math Methods in Physical Science
- MAT52-894 Senior Seminar in Mathematics (Capstone, to be taken in the fall of the senior year) or CSC54-894 Senior Seminar in Software Engineering (Capstone, to be taken in the spring of the senior year)

Major in Computational Mathematics (BS): 15 courses

- CSC54-184 Computer Science I
- CSC54-284 Computer Science II
- CSC54-384 Discrete Mathematics
- CSC54-394 Computer Organization
- CSC54-454 Algorithms
- MAT52-164 Modern Calculus I
- MAT52-264 Modern Calculus II
- MAT52-364 Modern Calculus III
- MAT52-674 Linear Algebra
- MAT52-754 Differential Equations
- Two courses from:
 - CSC54-414 Operations Research
 - CSC54-514 Database Management
 - MAT52-524 Introduction to Numerical Analysis
 - MAT52-574 Probability and Mathematical Statistics
- Two additional approved upper-level courses, in Mathematics or Computer Science or PHY53-454 Math Methods in Physical Science
- MAT52-894 Senior Seminar in Mathematics (Capstone, to be taken in the fall of the senior year) or CSC54-894 Senior Seminar in Software Engineering (Capstone, to be taken in the spring of the senior year)

Required supporting courses in the Computational Mathematics major (BS): 4 courses

- BIO50-173/171 Molecular and Cellular Foundations of Biology/lab, or BIO50-183/181 Evolutionary and Ecological Foundations of Biology/lab
- CHE51-103/101 General Chemistry I/Chemical Methods and Techniques Laboratory I
- PHY53-154 Fundamentals of Physics I
- One course from:
 - BIO50-183/181 Evolutionary and Ecological Foundations of Biology/lab, and BIO50-173/171 Molecular and Cellular Foundations of Biology/lab (counts as one course here since one is required above)
 - CHE51-203/201 General Chemistry II/Chemical Methods and Techniques Laboratory II
 - PHY53-164 Fundamentals of Physics II
 - PHY53-404 Electronics

See the Education Department for information regarding teacher certification in Mathematics and Computer Science.